One Cache, Two Workloads: Certifying KV Compression Along Pointwise and Aggregative Axes, with a Calibrated Expansion Gate

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Artifacts: `memsub-artifacts:/gate-eval/{eval-grid, ablation-results}.json`

Abstract

"Long context" conflates a serving problem—fitting the tokens—with two capability problems that separate the moment full attention becomes unaffordable: *pointwise* recall of exact, low-redundancy facts, and *aggregative* understanding of distributed, redundant content. We argue these are antagonistic optimization targets under any lossy cache compression, and we introduce a certification protocol that scores the two axes separately and forbids blending them. Applying the protocol to untrained KV eviction on Qwen3-4B over a ∼1B-token, four-domain corpus of 98,355 counterfactually filtered questions, we find the axes split almost completely: next-token KL against the full-context teacher stays near 3×10^{-2} while 0.45–0.55 of facts residing in evicted spans are silently lost (measurement noise floor: 0.000). We then show the pointwise axis can be restored by a *decoupled sufficiency gate*: a lightweight probe, trained for ≈$140 on counterfactual sufficiency labels produced by the frozen base model, that decides per query whether compressed memory suffices or exact re-expansion is required. The gate is natively calibrated (test ECE 0.017–0.023), strictly dominates a query-type-routing baseline at every matched expansion rate, and at its operating point cuts silent loss from 0.45–0.55 to 0.146 (bootstrap 95% CI $[0.136, 0.156]$; statistically indistinguishable from the 0.15 target) while expanding on only 16.8% of queries (oracle: 19.4%). A signal decomposition, however, shows the gate's discrimination (evicted-only AUC 0.622) is carried almost entirely by eviction-mechanics features rather than content embeddings—we therefore characterize the shipped gate as a *calibrated mechanics classifier* and identify content-aware sufficiency as open headroom. Two further findings refine the picture: on multi-hop questions the two-workload weakness appears on the *cost* axis rather than the correctness axis—the gate protects correctness by over-expanding 2.3× oracle—and gate discrimination transfers to held-out domains while calibration does not, making per-domain recalibration part of the definition of a deployable gate. Finally, we train a Still-class amortized synthesis compactor against the frozen base and certify it under the same protocol: at 1/40 of the published data recipe it does not displace untrained eviction on the aggregation axis (no-ship under our own rule), yet it *halves* silent needle loss (0.583 → 0.271), its recurrence collapses at depth 4, and the compactor-plus-refreshed-gate system reproduces the eviction-regime operating point with the credit assignment flipped: the compressor now does most of the protective work and the gate's marginal reduction falls from 72% to 22.5%—while the mechanics-vs-content signal decomposition *inverts*, making content the (weakly) better predictor under synthesis.

1 Introduction

Benchmarks, methods papers, and product claims routinely report a single "long-context" score. This paper's starting point is that no such single number can be coherent once the context

no longer fits in exact form. A needle-class query—an identifier, a figure, a verbatim clause—concentrates its answer in exact bytes with near-zero redundancy against the surrounding context; any lossy transform destroys it at a rate proportional to the compression ratio, and the optimal mechanism for it is selection, indexing, and exact lookup. An integration-class query—summarization, theme tracking, most document reasoning—lives on distributed, redundant information; it is compression-tolerant and often mildly compression-benefited. The two are not merely different but antagonistic: compression bought for the aggregative workload is paid for in pointwise fidelity, and exact retention kept for needles is memory that compression was meant to free.

If the antagonism is real, three things follow. Evaluation must report the two axes separately, because a blended score lets a method win the aggregate while being needle-dead. Compressed-memory systems need a *routed two-path architecture*—a lossy tier for aggregation and a lossless raw path for pointwise traffic—rather than a single representation with a quality knob. And the router itself becomes the critical learned component: a calibrated, per-query estimate of whether the compressed representation suffices.

We contribute, in order: **(i)** a two-axis certification protocol for compressed LLM memory, with counterfactually filtered probes such that a correct answer proves the information flowed through the (compressed) cache; **(ii)** a measurement at ∼1B-token scale across four document domains showing the axes split under untrained KV eviction—aggregative fidelity essentially intact, pointwise fidelity collapsing, the gap widening with eviction fraction; **(iii)** a decoupled sufficiency gate—trained entirely outside the frozen base model for ≈$140—that restores the pointwise axis at a fifth of the always-expand cost, with calibration good enough to expose as a probability, together with an ablation showing its signal is predominantly eviction mechanics rather than content; **(iv)** two phenomena we believe are documented here for the first time: the multi-hop *cost overhang*, where a well-calibrated single-hop gate protects multi-hop correctness only by systematically over-expanding, and *calibration non-transfer*, where discrimination generalizes across domains but calibrated probabilities do not; **(v)** a small-scale but fully-certified comparison of trained amortized synthesis against untrained eviction under the identical protocol, showing the two compression families fail *differently*: eviction deletes needles that synthesis partially preserves, synthesis degrades the aggregation axis that eviction leaves nearly intact at matched budget, and the sufficiency signal a gate can learn changes family with the compressor.

2 Related work

KV-cache reduction. Per-context selection scores tokens at inference and evicts the rest (H2O [12], SnapKV [6], StreamingLLM [10]); amortized selection learns the scorer (KV-Distill [1]); per-context and amortized synthesis write new compact entries rather than keeping original rows (Still [7]); architecture-level lines bake reduction into the model (MLA [2], native sparse attention [11]). Our work is orthogonal to the choice of compressor: we certify compressors along two axes and supply the decision layer their lossiness requires. Notably, the strongest published synthesis results themselves show the axis split we formalize—near-full aggregative utility with needle-retrieval collapse—which we take as independent evidence for the two-workload framing.

Selective prediction and abstention. The gate is a selective-prediction head [3] over an external memory: it estimates, per query, whether the model's conditioning state supports the answer. It relates to answerability and abstention heads in retrieval-augmented settings, with

two differences: the label is *counterfactual* (answers under compressed vs. raw conditioning are compared, rather than correctness being judged in isolation), and the head is trained fully decoupled from the backbone, using the backbone only offline as a teacher. Calibration machinery follows Guo et al. [4] and Platt [9].

Memory-augmented agents. OS-style memory hierarchies (MemGPT-class [8]) page text in and out of context but do not certify fidelity or route by workload; our protocol and gate are directly applicable to such systems. Blended long-context suites such as RULER [5] are exactly the evaluations our protocol unbundles.

3 Two axes and a certification protocol

Definitions. Fix a base model f, a compression policy C, and a corpus of (context, question, answer) items where each question is generated from a specific source span. An item is *counterfactually valid* if f answers it correctly with the context and incorrectly without it—so a correct answer under compression proves the information traversed the compressed cache. The **aggregation axis** is measured over integrative behavior: next-token KL between f conditioned on the full cache and f conditioned on C(cache), plus answer-accuracy deltas on items whose source spans survive compression. The **pointwise axis** is the *lost-fact incident rate*: among counterfactually valid items whose source span was compressed away, the fraction answered incorrectly *without* any expansion mechanism—the silent-failure rate.

Protocol. A configuration is certified per axis, never as a blend; the noise floor is established by running the identical probe battery at zero compression; and a configuration certified only on the aggregation axis may ship only behind an active, calibrated expansion gate for pointwise traffic. The last clause converts the antagonism from an observation into a deployment rule. Every report object embeds an evaluation-protocol record: the exact item partition entering each axis, the contents of the compacted context during probing (surviving KV only; no summary text), confirmation that expansion was disabled, and the scoring rule (argmax over answer-letter logits, greedy—format failures are structurally impossible).

4 The axes split: measurement at 1B-token scale

Setup. Base model Qwen3-4B-Instruct-2507 (bf16, greedy, pinned runtime fingerprint). Corpus: 40,000 long documents across financial filings (238M tokens), literature (416M), case law (119M), and source code (154M); 98,355 extractive multiple-choice items generated by the base model from random ∼1k-token sub-chunks, teacher-verified answerable with context and discarded if answerable without (98.4% of the 100k target; code has the lowest filter survival). Equivalence is answer-letter exact match; a manual audit of the complete pointwise-failure population (14 unique items) found 0% format noise, ≈80% genuine forgetting, and 2–3 flawed items. Compression policy: evict-oldest-non-sink at fractions 0.25 and 0.40, with the first four positions pinned as attention sinks [10]. Contexts are 4096-token per-item prefills with item placement randomized by hash, giving natural exposure to eviction.

Result. The zero-eviction control gives a lost-fact noise floor of exactly 0.000 (conditioned on prior-correct probes; the raw metric carries a 0.167 never-knew floor, which is why all headline

rates below condition on prior-correct). Under eviction the aggregation axis barely moves—mean KL 0.0303 (f=0.25) and 0.0339 (f=0.40) against the full-context teacher, answer deltas on surviving-span items ≈0—while the pointwise axis collapses: **0.453** (f=0.25, n=1,606) and **0.551** (f=0.40, n=3,493) of facts in evicted spans are silently lost. A matched-random control (same token count evicted uniformly rather than contiguously) inverts the damage profile—4× less pointwise loss at 3× the aggregation KL—showing that recency-structured policies concentrate pointwise damage exactly where aggregate metrics cannot see it. One artifact, two workloads, opposite outcomes.

(Protocol note: an earlier certification pass under packed multi-item contexts with needle-anchored eviction gave 0.708/0.808 *raw; the canonical numbers above are re-stated under the per-item randomized-placement protocol used throughout this paper, with the reconciliation in Appendix B.)*

5 A decoupled sufficiency gate

5.1 Labels, model, calibration

For each of the 98,355 items and each eviction fraction we produce a counterfactual sufficiency label: $y = 1$ iff the answer under the compacted cache differs from the answer under the raw span (196,710 rows; items whose source span survived eviction enter as natural y=0 rows). Splits are grouped by source document, domain-stratified, with an additional leave-one-domain-out rotation. The gate is a dual encoder over frozen Qwen3-Embedding-0.6B—the same embedding model that produces the memory blocks' retrieval shadows—with a 256-d normalized projection interaction, a small feature head (eviction fraction, query–block overlap, evicted flag, domain), and an auxiliary query-workload head. A frozen-embedder linear probe sufficed. Per-fraction temperature search returned T=1.0 for both fractions: the probe is natively calibrated, with test ECE 0.023 (f=0.25) and 0.017 (f=0.40). Marginal cost of the entire gate experiment—labels, training, evaluation, figures—was ≈$140.

5.2 What the gate learned

Pooled test AUC is 0.829, but pooling conflates two tasks: separating evicted from survived rows (near-deterministic given the eviction features) and the question that matters—among evicted spans, which lost their fact. Restricting evaluation to evicted-span rows, the full model reaches AUC **0.622** (95% CI $[0.608, 0.636]$; PR-AUC 0.621)—but the features-only ablation (both encoders zeroed) reaches 0.621 ± 0.006, within seed noise of the full model's 0.624 ± 0.004, while embeddings-only falls to 0.536 ± 0.004 (Table 1). The learned signal is predominantly eviction mechanics (fraction, overlap, position), not content semantics; this also explains the near-tie between document-level and item-level splits observed during training (doc-split AUC 0.8283 vs. leaky item-split 0.8259—there is little content signal for item leakage to exploit). We therefore describe the shipped gate as a *calibrated mechanics classifier* and treat content-aware sufficiency as open headroom.

5.3 Operating point and baselines

At the operating threshold τ=0.488 (chosen on the calibration split), the gate reaches test lost-fact incident rate 0.146 (bootstrap 95% CI $[0.136, 0.156]$ over test items; across 3 training seeds

Table 1: Signal decomposition (3 seeds, mean ± SD). Features-only matches the full model on the evicted-only task; embeddings add essentially nothing.

Variant	evicted-only AUC	evicted-only PR-AUC	pooled AUC
full model	0.624 ± 0.004	0.622	0.831
features-only (encoders zeroed)	0.621 ± 0.006	0.620	0.837
embeddings-only (features zeroed)	0.536 ± 0.004	0.561	0.557

Table 2: Matched-expansion-rate comparison (test split, lost-fact incident rate conditioned on prior-correct).

Expansion rate	0.1	0.2	0.3	0.4	0.5
gate	0.277	0.092	0.000	0.000	0.000
query-type routing	0.470	0.416	0.368	0.308	0.248

0.126 ± 0.014) at expansion rate 0.168 $[0.163, 0.173]$, against never-expand at 0.453–0.551 (per fraction), always-expand at rate 1.0, and the oracle at 0.000 loss / 0.194 rate. The upper confidence bound marginally exceeds the pre-registered 0.15 target, so we report the frontier rather than claim the threshold: the gate sits at 0.13–0.15 silent loss at ≈0.17 expansion. Against the interpretable baseline—classify the query as pointwise vs. aggregative and expand iff pointwise—the gate strictly dominates at every matched expansion rate from 0.1 to 0.5 (Table 2); the full cost–correctness plane is Figure 1, where the gate's curve lies inside the routing baseline everywhere and reaches ≈0 loss at ∼0.30 expansion rate.

5.4 Multi-hop: the weakness moves to the cost axis

On a 500-item slice of teacher-filtered two-fact questions composed within single documents, the gate at the single-hop threshold *holds correctness*—lost-fact incident rate 0.036 vs. 0.376 never-expand—but does so by over-expanding: recall 0.904 at precision 0.388, expansion rate 0.438 against an oracle of 0.188 (2.3×). The predicted mid-spectrum weakness of retrieval-then-integration queries is real, but it lands on cost, not correctness: a calibrated single-hop gate defends multi-hop questions by firing too often. This cleanly scopes the next component—a selector that scores block *sets* so a multi-hop query triggers one grouped expansion—and gives it a quantitative target (recover ∼2.3× → ∼1× oracle cost at held recall).

5.5 Calibration does not transfer; discrimination does

Under leave-one-domain-out evaluation the gate's discrimination holds (pooled AUC 0.781–0.865; near-zero lost-fact incident rate at ≈0.26 expansion), but calibration degrades to ECE 0.05–0.13. Fitting on ∼200 labeled items from the new domain repairs it—with a caveat the ablation surfaced: a scalar temperature alone left three of four domains above the 0.05 bar (0.071–0.073); the cross-domain miscalibration has a location component, and escalating to Platt scaling (temperature + bias on frozen scores) restores ECE to 0.014–0.035 in every rotation (Table 3). The practical statement: a "calibrated gate" is not a checkpoint but a checkpoint plus a per-domain recalibration procedure, and deployment to a new domain carries a small, explicit labeling cost.

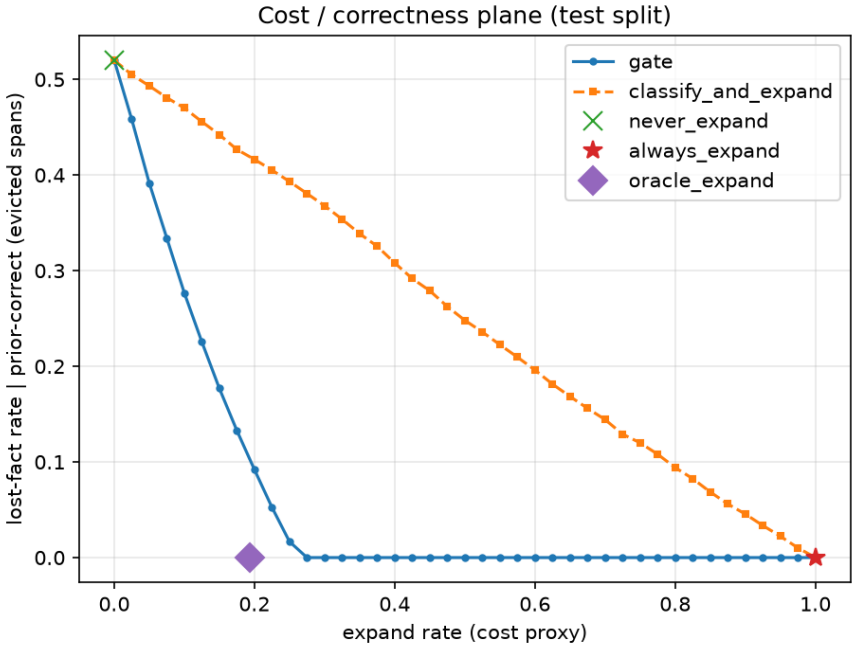


Figure 1: Cost–correctness plane (test split). The gate curve dominates query-type routing everywhere; oracle at $(0.194, 0)$; never-expand at $(0, 0.52)$ pooled.

Table 3: Leave-one-domain-out recalibration on ∼200 items from the held-out domain (frozen gate scores).

Held-out domain	method	ECE before	ECE after
financial	Platt	0.130	0.014
legal	Platt	0.106	0.029
code	temperature	0.064	0.035
literature	Platt	0.116	0.023

6 Does trained synthesis change the verdict?

The preceding sections certify *untrained* eviction. We next train a Still-class amortized compactor—one Perceiver per transformer layer (per-KV-head latent banks, d_l=256, two pre-norm blocks with QK-normed cross-attention at logit scale d_l, no FFN, ≈50M parameters), with the RoPE-fix (keys inverse-rotated to a position-free frame, compactor-internal RoPE at base 10, compact keys re-rotated at evenly spaced output positions)—against the frozen base, and certify it under the identical protocol. Two sanity gates precede training: an identity-initialization pass-through test (with one latent per position, compact-cache logits must match full-cache logits *before* training; this is the cheap detector for the position-metadata failure class) and a loss-sanity bar (forward KL on rationale-trace answer tokens, top-200 teacher support, must fall below 0.4 within 300 steps). Both passed. Training used ≈25k teacher-rationale triplets at 8k context for 1,500 steps at effective batch 32—about 1/40 of the published recipe's data volume—so the certification below judges *our checkpoint*, not the method's ceiling.

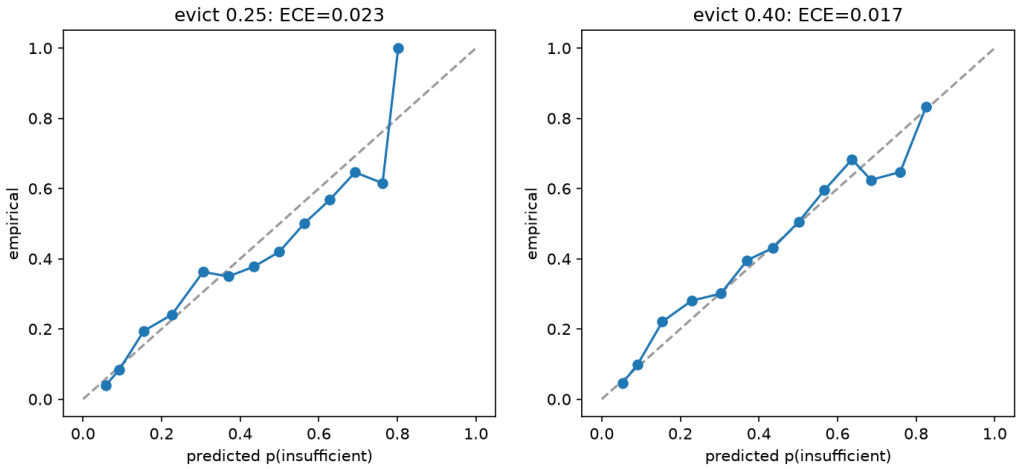


Figure 2: Reliability curves per eviction fraction; the linear probe is natively calibrated (T=1.0; ECE 0.023/0.017).

Table 4: Trained synthesis vs. untrained eviction at matched retained budget (sinks + t=1024 latents + 1,024 recent raw slots of a 4,096-token context; 12 contexts across 4 domains).

arm	mean KL	acc. delta	lost-fact | prior-correct
control (no compression)	0.000	0.000	0.000
eviction @ matched budget	**0.122**	−0.049	0.583
compactor, depth 1	0.263	−0.090	**0.271**
compactor, depth 4 (recurrence)	2.292	−0.056	0.750

No-ship on the aggregation axis; a surprise on the pointwise axis. Table 4: at matched budget our compactor is *worse* than eviction on the aggregation axis (KL 0.263 vs 0.122), so under our own deployment rule it does not replace eviction as the default compressed tier. But the pointwise axis inverts: synthesis *halves* silent needle loss (0.271 vs 0.583)—compact latents are lossy blends that still carry needle content eviction deletes outright. Both numbers sit far above the 0.000 floor, so the gate-mandatory rule re-attaches under every compaction regime measured. The two compression families fail *differently*, which strengthens the case for certifying both axes rather than ranking compressors on either alone.

Recurrence collapses. Certifying at deployment recurrence depth (re-compacting accumulated state chunk by chunk), depth 4 degrades KL by 9× (2.29) with 0.75 needle loss—consistent with the depth-validity envelope reported for Still-class compactors. We emit `max_certified_depth` = 0 into the compaction policy version (weights hash included), and resume paths refuse to stack compactions past it.

The gate signal changes family with the compressor—and so does the credit assignment. Relabeling all 98,355 items under the compactor policy (old prefix synthesized, recent quarter raw; positive rate drops to 0.167) and retraining the gate, with baselines re-derived *under this policy* rather than borrowed from the eviction frame: never-expand sits at 0.189, the

refreshed gate at 0.146 @ expansion 0.147 (ECE 0.007), the oracle at 0.000 @ 0.170. The resulting system point numerically matches the eviction-regime operating point (0.146 @ 0.168)—but the mechanism is different. Under eviction the gate cuts silent loss by 72% (0.520 → 0.146); under synthesis the *compressor* does most of the protective work and the gate adds a 22.5% marginal reduction while consuming 86% of the oracle's expansion budget. Consistent with that, the §5.2 decomposition *inverts*: embeddings-only (0.567 ± 0.005) now beats features-only (0.537 ± 0.002; full 0.554 ± 0.003)—there is no eviction window for mechanics features to read, content becomes the better predictor, and both are weak. The two regimes' evicted-only AUCs (0.62 eviction, 0.55 synthesis) bound what any downstream selection can do: set-scoring grouped expansion (rank-then-cut at the largest marginal-score gap, plus a trained joint re-scorer) reduces the multi-hop cost overhang only from 2.33× to 2.15–2.24× oracle. The overhang is not a grouping problem; it is the content-signal gap, measured three ways.

7 Limitations

Single base model and scale (4B). Our trained-synthesis comparison (§6) uses 1/40 of the published data recipe; the no-ship verdict certifies our checkpoint and budget, not the method—a full-recipe run could reverse the aggregation-axis ordering (though the depth-collapse and needle findings match the published pattern and should be robust). Extractive MCQ probes with a letter-match equivalence function rather than open-ended generation; greedy decoding; per-item single-turn contexts rather than multi-turn threads; multi-hop limited to k=2 within-document composition. The dead content pathway of §5.2 may be a gloss-quality artifact (extractive chunk prefixes and question-only query encodings) rather than a fundamental limit; probing that requires summary-grade glosses and is left open. The two-axis protocol itself is compressor- and model-agnostic; the specific numbers are not.

8 Conclusion

Single-number long-context evaluation is incoherent under compression: the pointwise and aggregative axes are antagonistic, and at 1B-token scale they split almost completely. The resolution is architectural—a lossy tier for aggregation, a lossless path for needles, and a router between them—and the router turns out to be cheap: a ≈$140 decoupled probe over frozen scores, natively calibrated, dominating query-type routing, and recovering the pointwise axis at a fifth of always-expand cost. The ablation adds the honest mechanism: today that router earns its keep from eviction mechanics, not content understanding, which bounds what it can know and marks content-aware sufficiency as the open problem. The trained-synthesis experiment sharpens the same conclusion from the other side: a compressor family that halves needle loss still cannot ship without the gate, the sufficiency signal changes family with the compressor, and both the gate's discrimination ceiling and the multi-hop cost overhang trace to one measured gap—content-aware sufficiency. That, plus per-domain recalibration as a shipped procedure, is a precise target rather than an open mystery, which is where we think a memory system's decision layer ought to leave its next iteration.

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A Evaluation-protocol record

Every certification report object embeds: (i) the item partition per axis—pointwise = needle items whose source chunk lies in the (reference) eviction window; aggregation = a disjoint same-document set, with the count of aggregation items whose chunks fall outside the certification context (they score ≈chance under both conditions and dilute, but cannot bias, the paired delta); (ii) compacted-context contents (surviving KV only; no summary text or auxiliary representation); (iii) expansion disabled during certification probing; (iv) scoring by argmax over answer-letter logits under greedy decoding, making letter-format failures structurally impossible; (v) the KL window length. All arms (control, per-fraction, matched-random) share one probe partition; identical prior-correct counts across arms (21/25) verify it.

B Baseline reconciliation

The earlier certification numbers (0.708/0.808 raw; 0.550/0.650 conditioned) and this paper's canonical never-expand rows (0.453/0.551) measure different protocols. (1) *Context construction*: the earlier pass certified 20 packed contexts whose eviction window was anchored on the needle chunk (adversarial placement; whole chunk always evicted); the canonical protocol uses per-item 4096-token contexts with hash-randomized placement, so evicted rows include boundary items with partial (> 0.5) overlap. (2) *Denominators*: the earlier raw rates include never-knew items (measured floor 0.167); canonical rows have raw-correct ≈1 by construction (generation-time filtering plus greedy determinism), so no never-knew inflation. (3) *Pooling*: the pooled 0.520 mixes both fractions. The paper uses only the canonical per-item protocol; the earlier numbers remain in the certification artifacts flagged as a different protocol.

C Decomposition, seeds, and uncertainty

Table 1 gives the three-variant decomposition (3 seeds each). GT9-class headline metrics across seeds: lost-fact incident rate 0.126 ± 0.014, expansion rate 0.177 ± 0.007. Bootstrap over test items (1,000 resamples): lost-fact incident rate $0.146\,[0.136, 0.156]$, expansion rate $0.168\,[0.163, 0.173]$, evicted-only AUC $0.622\,[0.608, 0.636]$.

D Corpus and filter survival

Four ungated sources (SEC 10-K filings; US court opinions; permissive-license source code; Project Gutenberg literature), 40,000 documents of $\geq \sim$6k tokens, ∼927M tokens total. Counterfactual filters retain ≈38% of generated items; per-domain yields 25,000/24,790/23,563/25,002 (financial/legal/code/literature). Legal-domain source text carries OCR noise, visible as the weakest per-domain gate recall.

E Reproducibility

Pinned environment (torch 2.7.1, transformers 4.55.2, safetensors 0.5.3); a 14-field compatibility fingerprint (model, tokenizer, chat template, RoPE config, attention implementation, KV layout, quantization, runtime, policy versions, kernel numeric probe) is stored with every KV artifact

and validated all-or-nothing. Compute ledger: corpus ≈$500; labels ≈$110; gate training, evaluation, multi-hop slice, and ablations ≈$30; trained-compactor experiment (rationale data prep, GT-gated 8×H100 training sweep with an iterative family, certification, relabeling, gate refresh, set-scoring evaluation) ≈$715. Gate and compactor checkpoints, calibration records (including per-domain recalibration entries), compaction policy versions (weights hash + certified depth), evaluation grids, and all per-item predictions are stored as content-addressed artifacts.